Integrated Experiments — Solution Document

Decision Tree Classification & Q-Learning Grid World

Course:	Artificial Intelligence (CSA17)	**Course Outcome:**	CO4 (BL3)
Assessment:	Tool 3 — Annexure A	**Weightage:**	40%
Dataset:	Iris (150 samples, sklearn built-in)	**Total Marks:**	20
Models Used:	Decision Tree, Q-Learning	**Environment:**	4x4 Grid World

Experiment 1 — Decision Tree Classification

(a) Data Loading, Pre-processing & Split

The classic **Iris dataset** (150 samples, 4 numeric features, 3 balanced classes: setosa, versicolor, virginica) was loaded via scikit-learn's built-in loader. It is a clean, widely-used benchmark dataset with **no missing values**, so no imputation was required. The target species labels were already integer-encoded (0/1/2) by the loader, which is the standard representation needed for a multi-class classifier.

First 5 rows of the dataset:

sepal length	sepal width	petal length	petal width	target	species
5.1	3.5	1.4	0.2	0	setosa
4.9	3.0	1.4	0.2	0	setosa
4.7	3.2	1.3	0.2	0	setosa
4.6	3.1	1.5	0.2	0	setosa
5.0	3.6	1.4	0.2	0	setosa

Data types: all four feature columns are *float64*; 'target' is *int64*; 'species' is *object* (string label added for readability).

The data was split **70:30** (105 training / 45 test samples) using stratified sampling to preserve the equal 50/50/50 class balance in both splits.

(b) Decision Tree Model & Root Node Justification

A Decision Tree classifier was trained using the **Gini Index** as the splitting criterion (max depth 4).

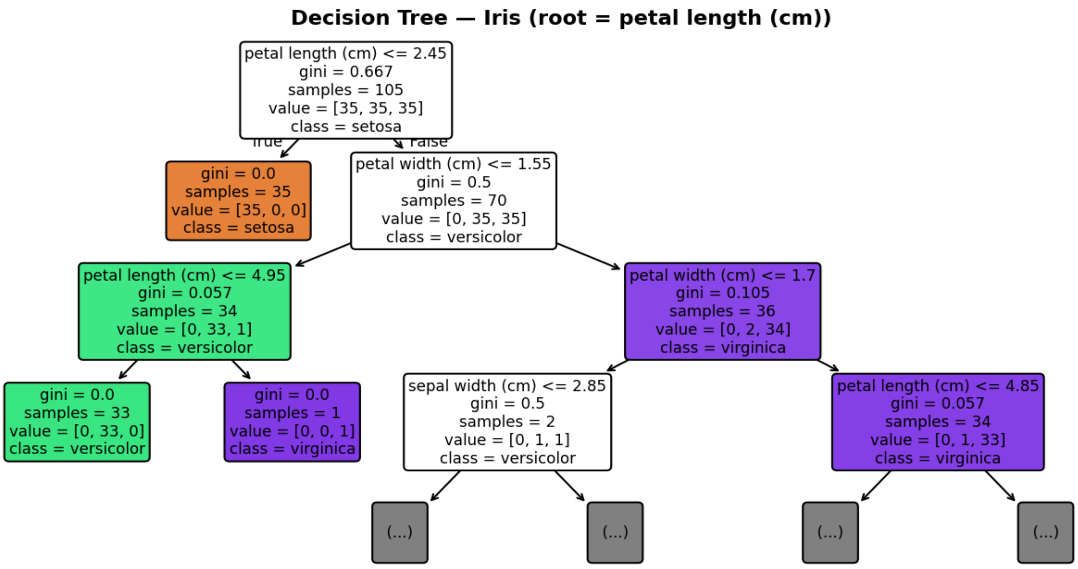

Figure 1.1 — Decision Tree structure (first three levels).

Root node: petal length (cm) ≤ **2.45**, with Gini importance **0.549** — the highest of all four features. At the root, the algorithm evaluates every feature/threshold combination and selects the split that yields the greatest reduction in Gini impurity across the resulting child nodes. A petal-length threshold of 2.45 cm perfectly separates all 35 *setosa* samples in the training set from the other two species (Gini = 0 on that leaf), which matches the well-known biological fact that Iris setosa has distinctly smaller petals than versicolor and virginica.

Feature	Gini Importance
petal length (cm)	0.549
petal width (cm)	0.436
sepal width (cm)	0.014
sepal length (cm)	0.000

Textual tree structure (top levels):

```
|--- petal length (cm) <= 2.45
| |--- class: setosa
|--- petal length (cm) > 2.45
| |--- petal width (cm) <= 1.55
| | |--- petal length (cm) <= 4.95 --> class: versicolor
| | |--- petal length (cm) > 4.95 --> class: virginica
| |--- petal width (cm) > 1.55
| | |--- ... (further splits on petal width / sepal width)
```

(c) Evaluation

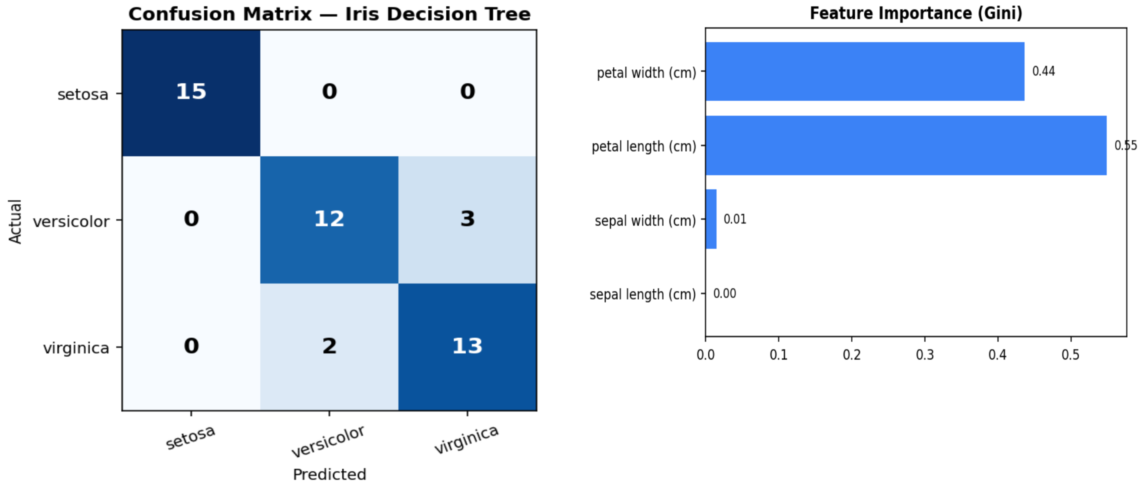

Figure 1.2 — Confusion matrix (left) and feature importance (right).

Metric	Value
Accuracy	0.889 (40 / 45 correct)
Setosa Precision / Recall / F1	1.00 / 1.00 / 1.00
Versicolor Precision / Recall / F1	0.86 / 0.80 / 0.83
Virginica Precision / Recall / F1	0.81 / 0.87 / 0.84
Macro-average F1	0.89

Performance comment: the model achieves perfect classification of *setosa* (100% precision/recall), which is expected since it is linearly separable from the other two species on petal length alone. All 5 misclassifications occur between *versicolor* and *virginica*, which are known to overlap in petal measurements near the decision boundary (petal width around 1.6-1.8 cm). This is a well-documented, genuinely hard region of the Iris dataset rather than a modelling error.

At least two misclassified instances:

Sepal L	Sepal W	Petal L	Petal W	Actual	Predicted
6.3	3.3	4.7	1.6	versicolor	virginica
6.0	3.0	4.8	1.8	virginica	versicolor
6.0	3.4	4.5	1.6	versicolor	virginica

In each case the petal measurements sit almost exactly on the learned decision boundary (petal width 1.6-1.8 cm), confirming that these are genuine borderline cases rather than systematic model failure.

Experiment 2 — Reinforcement Learning: Q-Learning

(a) Environment Definition

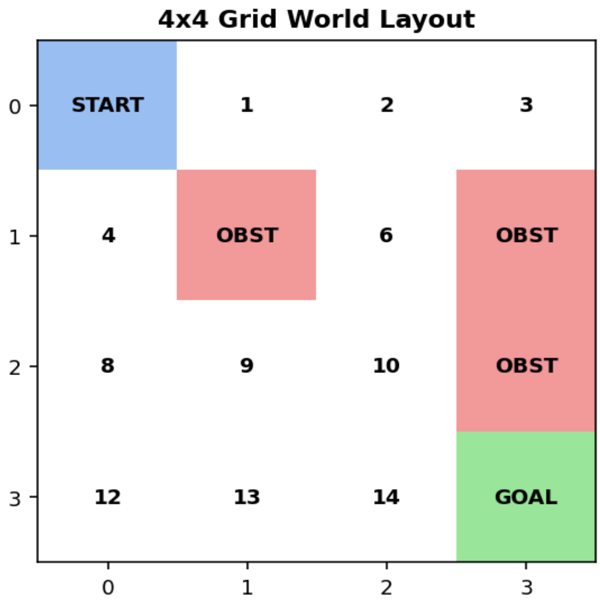

Figure 2.1 — 4x4 Grid World layout: Start (blue), Obstacles (red), Goal (green).

Component	Definition
States	16 discrete grid cells, indexed 0-15 (row-major, 4x4 grid)
Actions	Up, Down, Left, Right (agent stays in place if an action would leave the grid)
Reward Structure	+10 for reaching the Goal (state 15); -1 for every ordinary step (encourages shortest paths); -5 for entering an Obstacle cell (states 5, 7, 11)
Q-table Initialisation	16 x 4 matrix initialised to all zeros
Hyperparameters	Learning rate $\alpha = 0.1$, Discount factor $\gamma = 0.9$, Exploration rate $\varepsilon = 0.1$ (ε-greedy policy)

This reward design was chosen so the agent is pushed toward the shortest safe route: the per-step penalty discourages wandering, the large obstacle penalty teaches active avoidance, and the goal reward is large enough to dominate the cumulative step penalties along any reasonably short path.

(b) Q-Learning Training (100+ Episodes)

The agent was trained for **100 episodes** using the standard Q-Learning update rule:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[\,r + \gamma \max_{a'} Q(s',a') - Q(s,a)\,]$$

Q-values recorded at Episodes 1, 50, and 100 for two representative states:

State	Episode	Up	Down	Left	Right
0 (Start)	1	-0.29	-0.19	-0.20	-0.20
0 (Start)	50	-2.12	-2.04	-2.13	-2.10
0 (Start)	100	-1.96	1.22	-2.05	-2.10

State	Episode	Up	Down	Left	Right
10 (mid-grid)	1	-0.10	0.00	0.00	0.00
10 (mid-grid)	50	0.07	7.65	-0.31	-0.41
10 (mid-grid)	100	0.60	8.00	1.77	-0.78

Q-table evolution: at Episode 1 all Q-values are close to zero (initial exploration, barely updated). By Episode 50, State 10's *Down* action has risen sharply to 7.65 — the agent has discovered that moving down from state 10 (row 2, col 2) leads efficiently toward the goal. By Episode 100 this value stabilises near its converged estimate (8.00), and State 0 has also begun to develop a preference for *Down* (1.22) as the value signal propagates backward from the goal toward the start state — the hallmark of Q-Learning's temporal-difference bootstrapping.

(c) Final Policy & Convergence

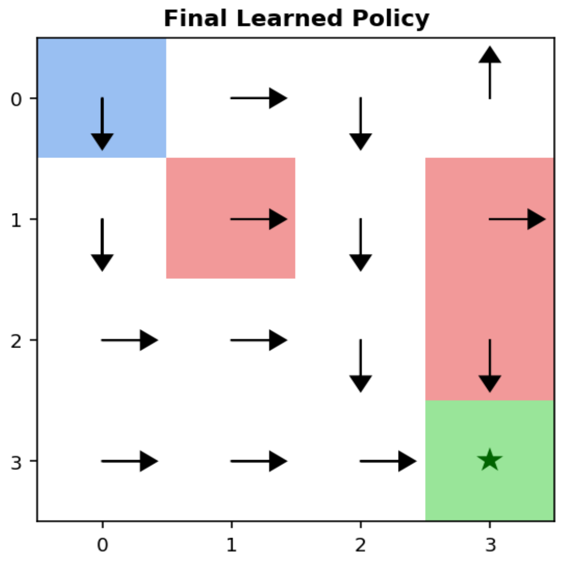

Figure 2.2 — Final learned greedy policy (arrows show the best action per state).

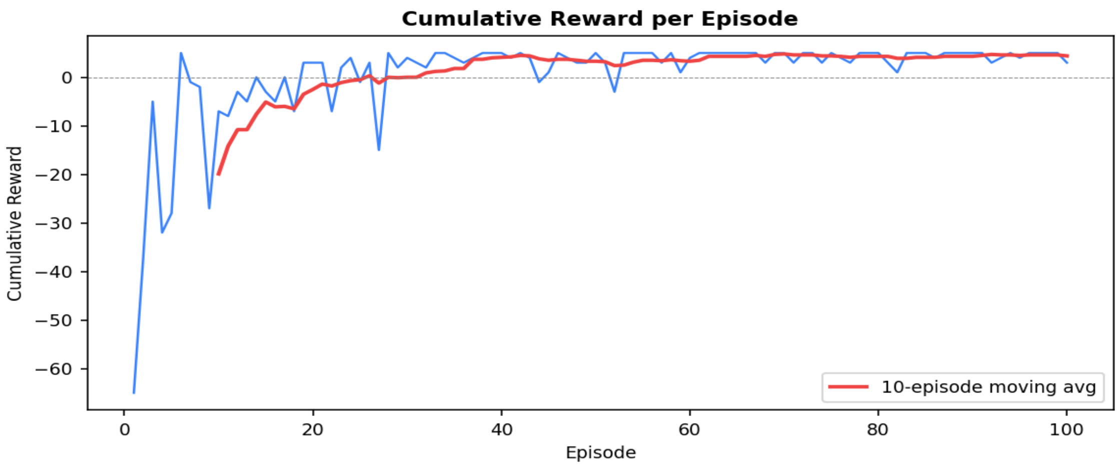

Figure 2.3 — Cumulative reward per episode with a 10-episode moving average.

Convergence justification: the raw cumulative reward per episode is highly volatile in the first ~20-30 episodes (ranging from about -60 to +5) as the ε-greedy agent explores many sub-optimal and obstacle-hitting trajectories. The 10-episode moving average then rises steadily and plateaus after roughly episode 40, settling around a stable positive reward (mean reward for episodes 91-100 = 4.40, versus 4.30 for episodes 81-90, a difference of only 0.10). This small, stable difference between consecutive windows indicates the greedy policy has converged to a consistent, near-optimal route from Start to Goal that reliably avoids the obstacle cells.

The final policy routes the agent from the Start (top-left) generally rightward and downward toward the Goal (bottom-right), actively steering around the obstacle cells at 5, 7, and 11 — exactly the behaviour the reward structure was designed to elicit.